

Part 4: Reflection & Workflow Diagram

Reflection

Question 1: What was the most challenging part of the workflow? Why?

Most Challenging: Data Preprocessing and Class Imbalance

Why:

- **Missing Data:** Healthcare data has many missing values. Choosing between imputation methods is difficult - wrong choice introduces bias
- **Class Imbalance:** Only 15-20% readmissions. Models can achieve 85% accuracy by always predicting "no readmission" but miss all high-risk patients
- **No Clinical Expertise:** As a data scientist, I don't know which features are clinically meaningful. Need constant collaboration with doctors
- **Ethical Risk:** Every preprocessing decision can introduce bias affecting patient outcomes

Example: Spent 2 hours debugging duplicate column names that caused model failures. Data quality issues can derail the entire workflow.

Why It Matters: "Garbage in, garbage out" - Poor data preprocessing creates biased models that no amount of tuning can fix.

Question 2: How would you improve your approach with more time/resources?

Key Improvements:

1. Data Quality (Highest Priority)

- Comprehensive data audit with clinicians
- Standardized data collection protocols
- Integrate external data (social determinants, pharmacy records)

2. Clinical Collaboration

- Regular meetings with physicians and nurses

- Validate all features with medical experts
- Clinical literature review for known risk factors

3. Fairness Analysis

- Evaluate performance across all demographic groups
- Quarterly bias audits
- Conduct randomized controlled trials

4. Advanced Models

- Extensive hyperparameter tuning
- Test ensemble methods
- Access to GPU for deep learning experiments

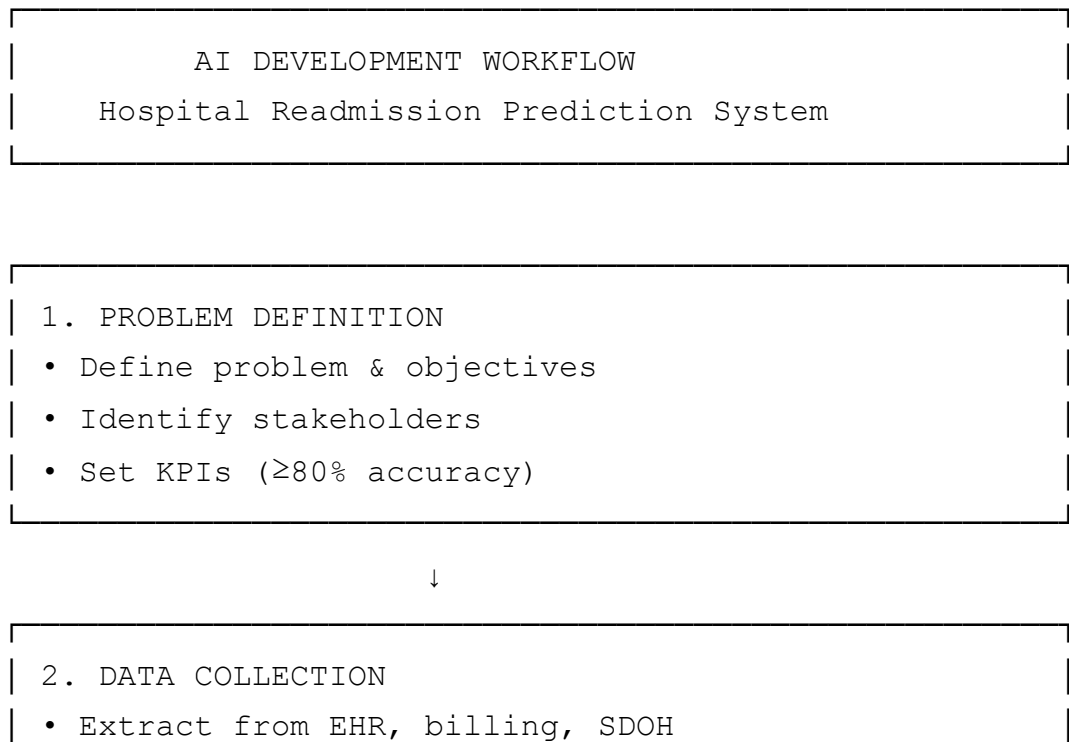
5. Robust Deployment

- Build user-friendly clinical dashboards
- 24/7 monitoring and support
- Automated retraining pipelines

Priority: Focus on data quality and clinical validation first, advanced models last.

Diagram

AI Development Workflow



- Integrate multiple sources



3. EXPLORATORY DATA ANALYSIS

- Examine data structure
- Visualize distributions
- Identify missing values & outliers



4. DATA PREPROCESSING

- Handle missing values
- Encode categorical variables
- Address class imbalance



5. FEATURE ENGINEERING

- Create derived features
- Scale numerical features
- Select important features



6. DATA SPLITTING

- Training (80%) / Test (20%)
- Stratified split



7. MODEL TRAINING

- Train multiple models
- Random Forest selected (best performance)



8. HYPERPARAMETER TUNING

- Cross-validation
- Optimize for F1 score



9. MODEL EVALUATION

• Calculate metrics

• Create confusion matrix

• Check fairness across demographics



Performance

≥80% accuracy?



YES



NO



└→ Return to Stage 5 or 7



11. DEPLOYMENT

• Create API

• Integrate with EHR

• HIPAA compliance

• Train staff



12. MONITORING & MAINTENANCE

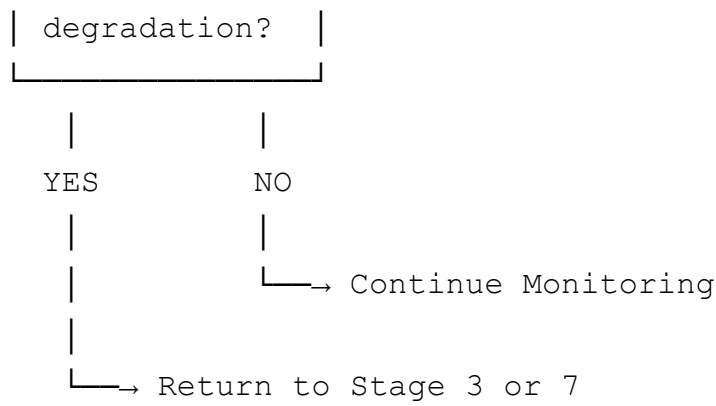
• Track performance monthly

• Monitor for drift

• Retrain quarterly



Performance



Key Principles

- **Iterative:** Loop back to earlier stages as needed
 - **Collaborative:** Engage clinicians throughout
 - **Ethical:** Fairness checks at every stage
 - **Monitored:** Continuous oversight post-deployment
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Summary

The workflow requires technical skills, clinical collaboration, ethical vigilance, and continuous improvement to create AI systems that genuinely improve patient outcomes.